Task 3: Requirements Architecture and Technology Setup

Name: Vedant Rechwad **Roll No.:** 23102A0054

1. Objective

Set up the Spring Boot application, dependencies, database configuration, project structure, and initial delivery-management architecture.

2. Spring Initializr

• Build: Maven

• Language: Java

• Spring Boot: 4.1.1

• Group: com.deliverypromise

• Artifact: delivery-promise-calculator

• Package: com.deliverypromise.calculator

• Packaging: Jar

• Java: 21

3. Dependencies

Spring Web/WebMVC, Thymeleaf, Spring Data JPA, Validation, DevTools, SQLite JDBC, and Hibernate Community Dialects.

4. SQLite POM additions

```
<dependency>
    <groupId>org.xerial</groupId>
    <artifactId>sqlite-jdbc</artifactId>
    <version>3.53.4.0</version>
</dependency>
<dependency>
    <groupId>org.hibernate.orm</groupId>
    <artifactId>hibernate-community-dialects</artifactId>
</dependency>
```

5. application.properties

```
spring.application.name=delivery-promise-calculator
spring.datasource.url=jdbc:sqlite:delivery_promise.db
spring.datasource.driver-class-name=org.sqlite.JDBC
spring.jpa.database-platform=org.hibernate.community.dialect.SQLiteDialect
spring.jpa.hibernate.ddl-auto=update
spring.jpa.show-sql=true
spring.jpa.properties.hibernate.jdbc.use_get_generated_keys=false
server.port=8765
```

6. Project Structure

```
src/main/java/com/deliverypromise/calculator/
■■■ DeliveryPromiseCalculatorApplication.java
■■■ HomeController.java
■■■ controller/DeliveryController.java
■■■ model/Delivery.java
■■■ repository/DeliveryRepository.java
■■■ service/DeliveryService.java

src/main/resources/
■■■ application.properties
■■■ templates/
    ■■■ index.html
    ■■■ deliveries.html
```

7. Application Logic

The Delivery entity stores customerName, deliveryAddress, productName, deliveryDays, promisedDate, and status. The service calculates promisedDate as LocalDate.now().plusDays(deliveryDays), defaults status to PENDING, and supports

customer/product search and status updates.

8. Commands

```
mvn clean test
mvn spring-boot:run
# Browser verification
http://localhost:8765/
```

9. Result

The application built and ran successfully, with SQLite persistence and the initial delivery-management pages available.

Task 3 — Actual Project Screenshots

Screenshots captured during the implementation are included below as evidence. These are the actual project screenshots, not recreated diagrams.

Figure 1. Spring Initializr configuration used to create the Java/Spring Boot project.



Figure 2. Java/Maven environment verification.

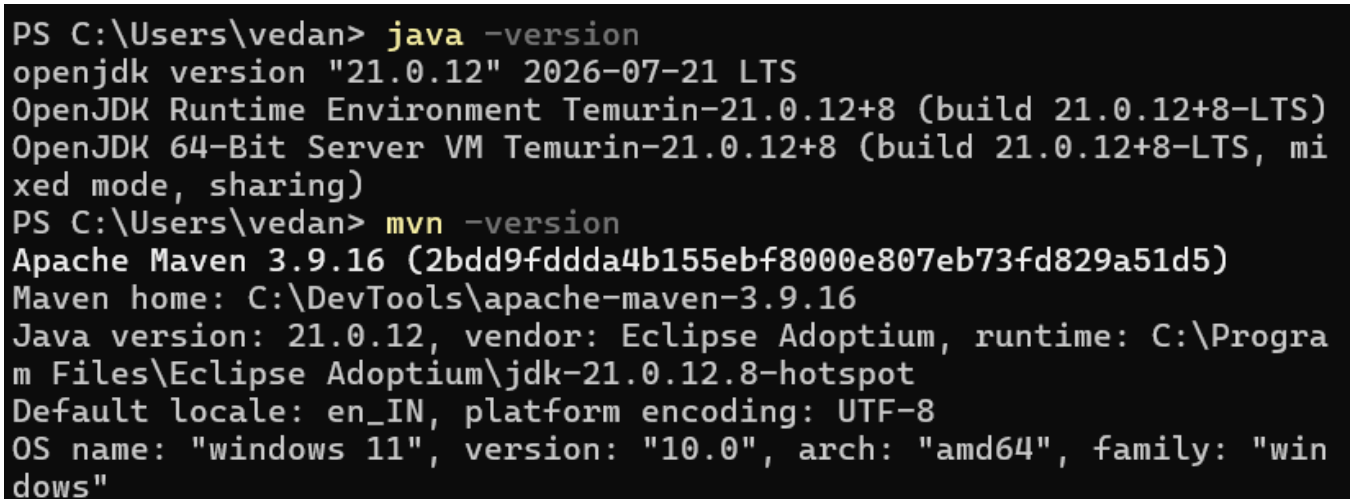


Figure 3. Running Delivery Promise Calculator on localhost:8765.

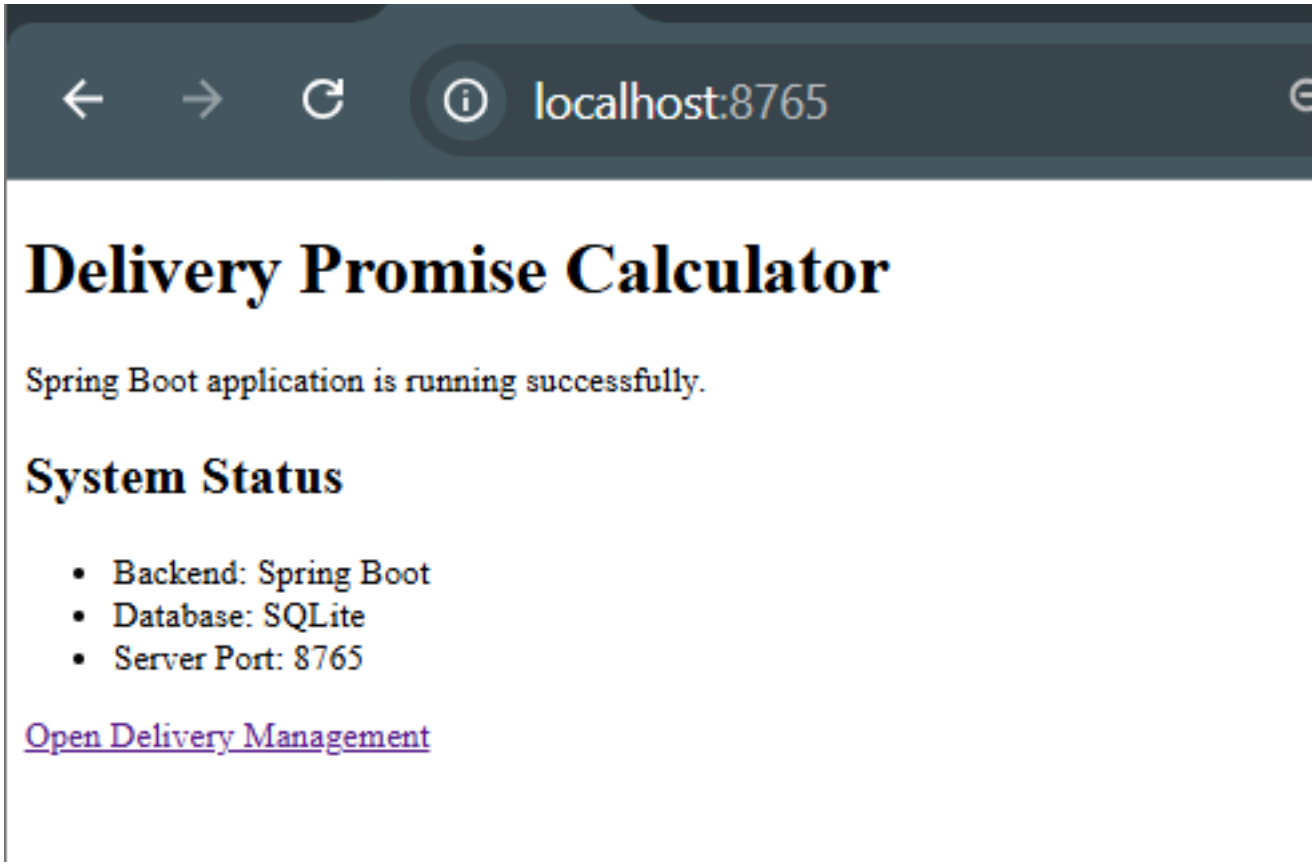


Figure 4. Maven build/package execution and successful build output.

